AGENTIC AI SYSTEMS · PAPER PRESENTATION

GAIA

A Benchmark for General AI Assistants

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2023 · arXiv:2311.12983

Roadmap

01 **Motivation**

02 Related Work

03 Research Question

04 The GAIA Benchmark

05 Results

06 Discussion

From Knowledge to AI Agents

2020–2024 · milestones in capabilities and evaluation

2020
MMLU
Knowledge benchmark

Oct 2022
ReAct
Reasoning + actions

Nov 2022
ChatGPT (GPT-3.5)
Conversational AI

Mar 2023
GPT-4 + Plugins
Browsing, code, APIs

Jun 2023
Function calling
Structured tool calls

Jul 2023
WebArena
Interactive web tasks

Aug 2023
AgentBench
Interactive agent eval

Oct 2023
SWE-bench
Real GitHub issues

Nov 2023
GAIA
General AI assistants

Apr 2024
OSWorld
Computer-use agents

GAIA: can assistants solve realistic, multi-step questions?

MOTIVATION

How Should We Evaluate AI Assistants?

BEFORE

LLMs as Chatbots

Answer from learned knowledge

Evaluation: MMLU

2023 · EMERGING AGENTS

LLMs + Tools

Browse · Reason · Compute

Evaluation: ?

Can we evaluate what AI can do, not just what it knows?

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Knowledge Benchmarks

BENCHMARK / YEAR	WHAT IT TESTS	GPT-4o	HUMAN
MMLU · 2020	Knowledge across 57 subjects	**88.7%**	**89.8%**
GSM8K · 2021	Multi-step grade-school math	**95.9%**	**60%**
GPQA · 2023	Graduate-level STEM questions	**53.6%**	**65%**

By 2024, some benchmarks neared saturation; models matched or exceeded certain human baselines.

MMLU — High School Biology

Which of the following is NOT involved in controlling cell division?

A Cyclins

B Protein kinases

C Checkpoints

D Fibroblast cells

Answer: D · Fibroblast cells are a cell type, not a cell-cycle control mechanism.

GSM8K — Grade-School Math

Janet’s ducks lay 16 eggs a day. She eats 3 and uses 4 for baking. She sells the rest at $2 per egg. How much does she earn daily?

Step 1 **16 − 3 − 4 = 9 eggs to sell**

Step 2 **9 × $2 = $18**

Final answer: $18

GSM8K tests multi-step mathematical reasoning from a self-contained word problem.

Interactive Agent Benchmarks

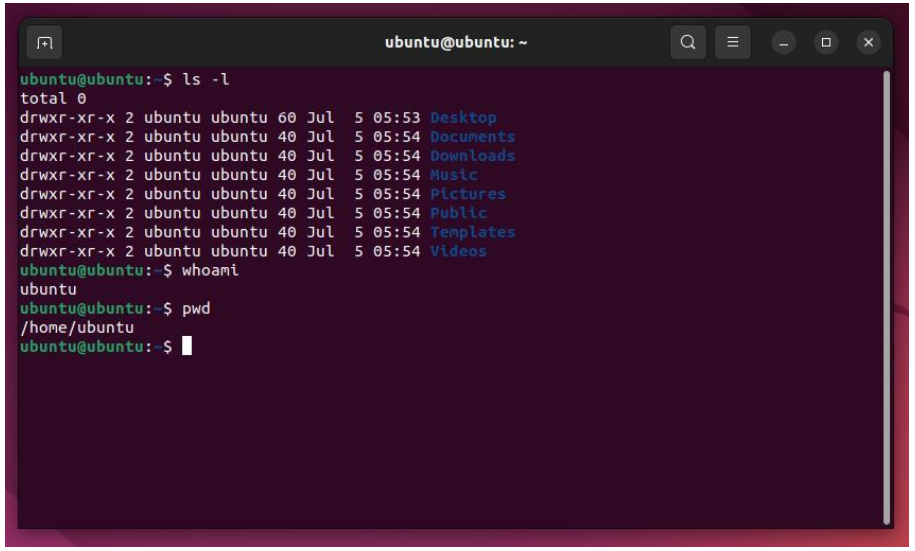
BENCHMARK / YEAR	ENVIRONMENT	GPT-4	HUMAN
AgentBench · 2023	OS · database · games · web	**4.01***	—
WebArena · 2023	Tasks on realistic websites	**14.41%**	**78.24%**

Agents interact through environment-specific actions (e.g., browser clicks or shell commands).

* AgentBench overall score (not a percentage). WebArena reports task success rate.

AgentBench — Operating System

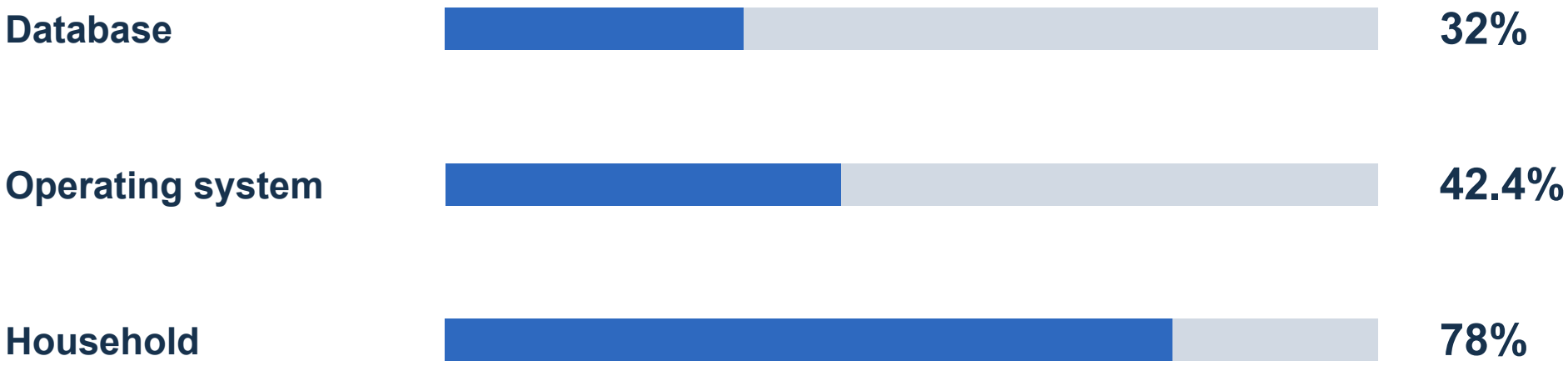
Find the number of Linux users who belong to at least two groups.



Inspect users · Check memberships · Count

One Model, Different Environments

GPT-4 · Success rate



Environment design can affect measured agent performance.
Building and maintaining new environments also limits scalability.

Specialized Task Benchmarks

BENCHMARK / YEAR	TASK	BEST BASELINE
SWE-bench · 2023	Fix real GitHub issues	**1.96%**

Repository + issue → Code patch → Tests

A real software task, but limited to a specific domain.

What Does Each Benchmark Fixes?

BENCHMARK	ENVIRONMENT	ACTIONS	EVALUATION
AgentBench	8 interactive environments	Environment-specific actions	Task success / reward
WebArena	Self-hosted websites	Click · type · navigate	Website state / answer
SWE-bench	GitHub repos + Docker tests	Agent-dependent; no fixed tool API	Patch passes tests
GAIA	Questions + associated files	Agent-dependent; no fixed tool API	Final answer correctness

What Does Each Benchmark Fix?

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GAIA fixes the questions and answers—not the tools or action interface.

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MAIN RESEARCH QUESTION

How Can We Evaluate General-Purpose AI Assistants?

Real-World Questions

Open web · Multimodality · Tool use
Not restricted to fabricated environments

Easy to Interpret

Conceptually simple · Human-solvable

Resistant to Shortcuts

Multiple steps · Curated questions
Reduced data contamination

Easy to Evaluate

Concise · Unambiguous answers

Secondary Research Questions

Once we have a benchmark, what can it tell us about AI assistants?

01 **How capable are AI assistants on real-world tasks?**

02 **How does task difficulty affect performance?**

03 **How much does tool access help?**

Example for GAIA Question

GAIA Question: The attached Excel file contains the sales of menu items for a local fast-food chain. What were the total sales that the chain made from food (not including drinks)? Express your answer in USD with two decimal places.



uploaded.xlsx

Example for GAIA Question

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uploaded.xlsx

Real world question: A practical sales-analysis request

The assistant must work with an attached Excel file.

Example for GAIA Question

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uploaded.xlsx

Easy to Interpret: A non-expert can understand the task, the agent needs to exclude drinks, then total food sales.

The difficulty is carrying out the steps, not understanding the goal.

Example for GAIA Question

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uploaded.xlsx

Resistant to Shortcuts: The answer depends on the spreadsheet's contents, not a memorized fact.

The assistant needs to inspect the file and calculate; this reduces, but cannot eliminate, shortcut risks.

Example for GAIA Question

GAIA Question: The attached Excel file contains the sales of menu items for a local fast-food chain. What were the total sales that the chain made from food (not including drinks)? Express your answer in USD with two decimal places.



uploaded.xlsx

Easy to Evaluate: The result is just amount

A concise, unambiguous answer makes correctness straightforward to check.

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Human-Curated Questions

01

Human creates

Question + one exact answer

→

02

Human documents

Solution trace, tools and time required

→

03

2 humans validate

Solve independently, check answer; repair/reject

466 curated questions

Example: A Human-Created GAIA Question

The authors wrote initial GAIA questions to guide human annotators.

Published annotated question

What was the actual enrollment count of the clinical trial on H. pylori in acne vulgaris patients from Jan–May 2018 as listed on the NIH website?

Answer: 90

The authors reviewed examples, but the paper does not identify this as one of them.

Instructions to Human Annotators

Grounding & novelty

01 Use a source of truth (Wikipedia, arXiv, GitHub, etc.); combine sources for Levels 2–3.

02 The answer must not already appear online in plain text.

03 The answer must remain stable over time, including if a source is deleted.

Answer & usefulness

04 Answer must be a number or at most a few words.

05 The answer must be unambiguous.

06 Question should be interesting: a useful assistant capability.

07 A human should solve it in reasonable time.

Human Solution Trace · All 8 Steps

Example: NIH clinical-trial enrollment question

01 Searched “nih” on Google search.

02 Clicked the top link to nih.gov.

03 Searched “h pylori acne” in the search box.

04 Clicked “More” and selected “Clinical Trials”.

05 Clicked the result about H. Pylori and acne.

06 Checked the date: January to May 2018.

07 Opened “Tabular View”.

08 Scrolled to Actual Enrollment and recorded 90.

Answer: 90 | 8 steps | Web browser (1 tool) | 8 minutes

2. Validate & Curate

Two independent annotators solve each question.

Agree

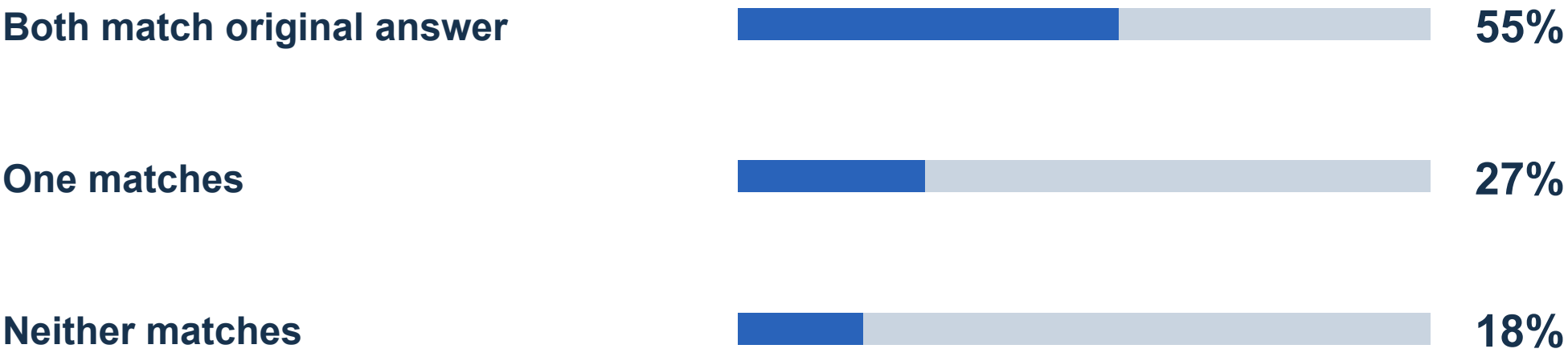
Disagree

Unclear

Goal: reliable, easy-to-check ground truth—not just difficult questions.

From 623 Questions to a Reliable Dataset

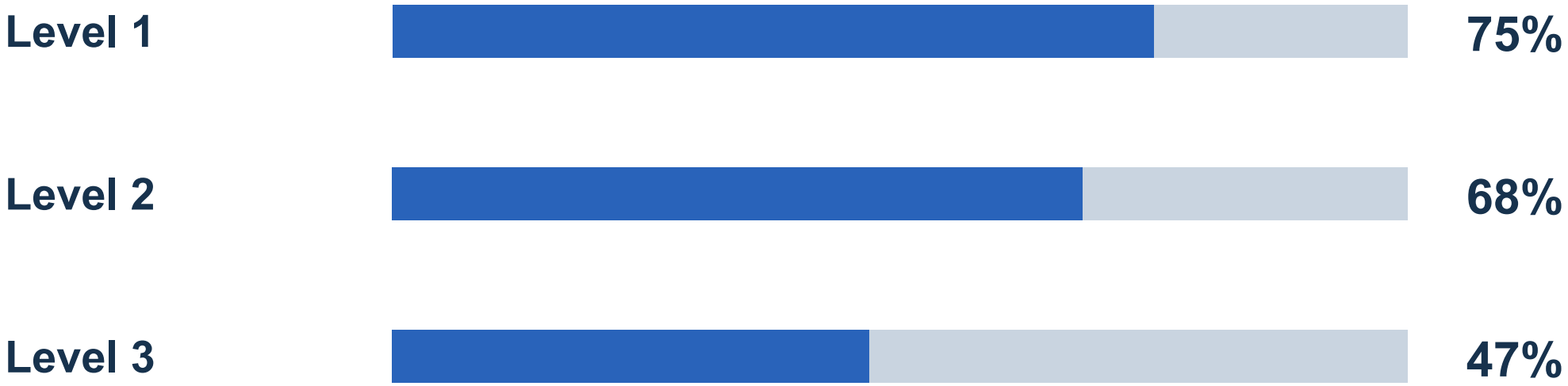
623 new questions · 1,246 independent validation attempts



68% judged valid after review · Final GAIA benchmark: 466 questions

Harder Questions Are Harder to Validate

Share of candidate questions judged valid, by difficulty level



Only 47% of Level 3 candidate questions passed validation.

Three Levels of Increasing Difficulty

Before: 623 newly crafted candidates | Final dataset: 466 questions

Level	Typical effort	Validated	Final count
1	≤5 steps · 0–1 tool	**75%**	**146**
2	~5–10 steps · multiple tools	**68%**	**245**
3	Long action chains · any tools	**47%**	**75**

Level 1 · Find a Fact

Question

What was the actual enrollment count of the clinical trial on H. pylori in acne vulgaris patients from Jan–May 2018 as listed on the NIH website?

Required: identify the NIH study and read its enrollment field.

Ground truth: 90

Level 2 · Combine Image + Web

Question

If this whole pint is made up of ice cream, how many percent above or below the US federal standards for butterfat content is it, using the standards reported by Wikipedia in 2020?

Answer as + or − a number, rounded to one decimal place.

Inputs: attached photo of a pint's nutrition label + Wikipedia

Ground truth: +4.6

Level 3 · Multi-Hop Research

Question (paraphrased for readability)

NASA’s Astronomy Picture of the Day (Jan 21, 2006) shows two astronauts. Identify the smaller astronaut and his NASA Astronaut Group. Among group members who flew in space, who spent the least total time in space, as of Aug 2023?

Return the astronaut's last name and total minutes in space.

Image interpretation → identify group → compare spaceflight records

Ground truth: White; 5,876

How Is GAIA Evaluated?

1

Question

Natural language prompt
+ optional attached file

→

2

Assistant

Any reasoning, browsing,
code or other available tools

→

3

Final Answer

Short number, phrase,
or comma-separated list

Only the final answer is scored.

The Evaluation System Prompt

System prompt: You are a general AI assistant. I will ask you a question. Report your thoughts, and finish your answer with the following template: FINAL ANSWER: [YOUR FINAL ANSWER].
YOUR FINAL ANSWER should be a number OR as few words as possible OR a comma separated list of numbers and/or strings.
If you are asked for a number, don't use comma to write your number neither use units such as $ or percent sign unless specified otherwise.
If you are asked for a string, don't use articles, neither abbreviations (e.g. for cities), and write the digits in plain text unless specified otherwise.
If you are asked for a comma separated list, apply the above rules depending of whether the element to be put in the list is a number or a string.

GAIA Question: Your final answer should be a number, or as few words as possible. Where were the Vietnamese specimens described by Kuznetzov in Nedoshivina's 2010 paper eventually deposited? Just give me the city name without abbreviations.

Evaluation Metric · Accuracy

Quasi-exact match

Compare the final answer with the ground truth after type-dependent normalization.

Number	Ignore inconsequential formatting such as currency symbols and separators
Short text	Normalize case, punctuation and whitespace
List	Compare the expected items in order

Accuracy = correctly matched final answers / evaluated questions

Human Baseline · Independent Solvers

92% **accuracy overall**

Computed from the independent validators' attempts on questions judged valid—not the original question authors.

Level 1	Level 2	Level 3
94%	**92%**	**87%**

Search Engine Baseline

A human checks whether a simple search reveals the answer.

GAIA question → **search engine** → **First results page**

A human submits the full GAIA question to a search engine, inspects the first page of results, and judges whether the correct answer can be deduced.

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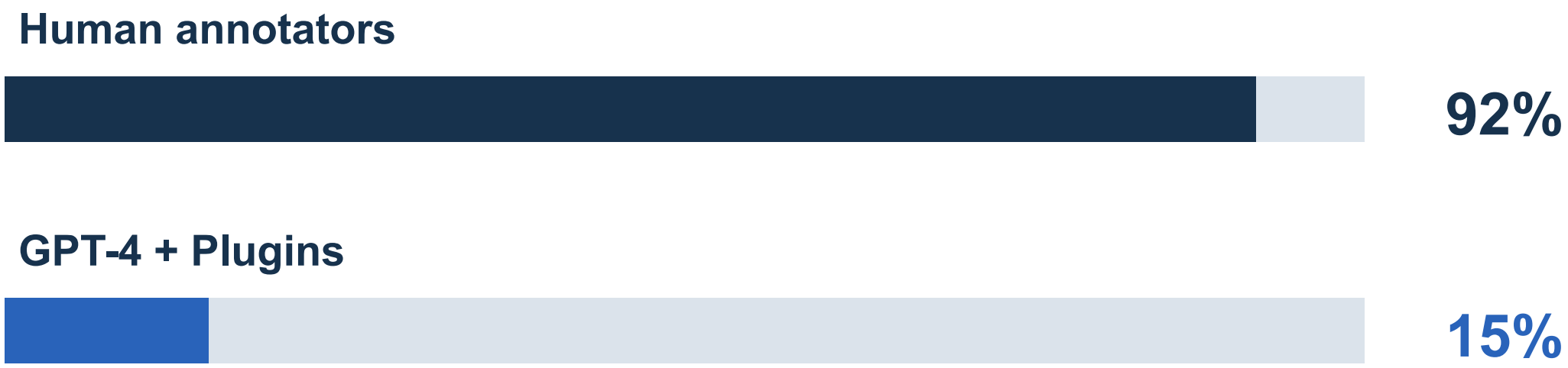
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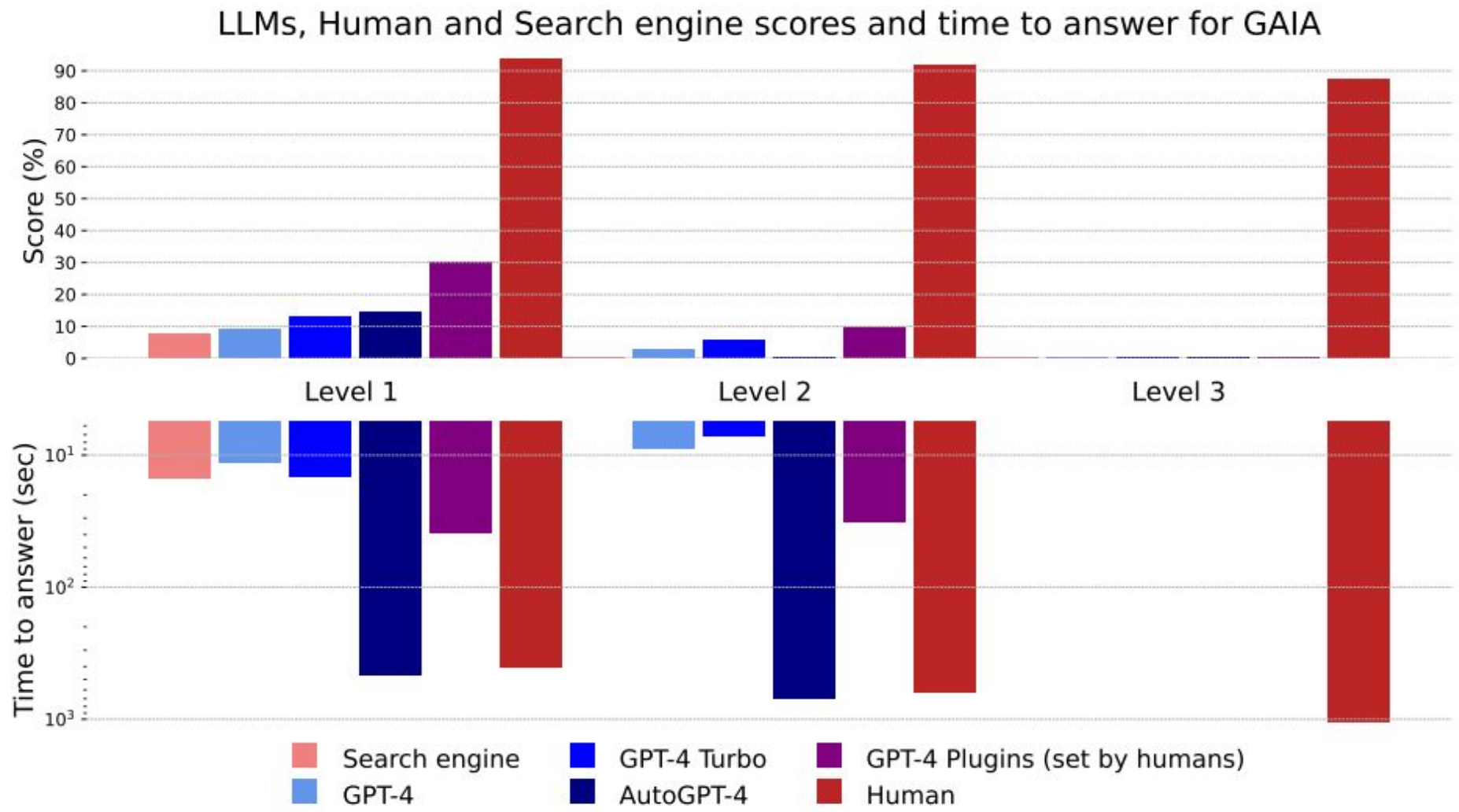
06 Discussion

The Main Result: Humans vs. AI

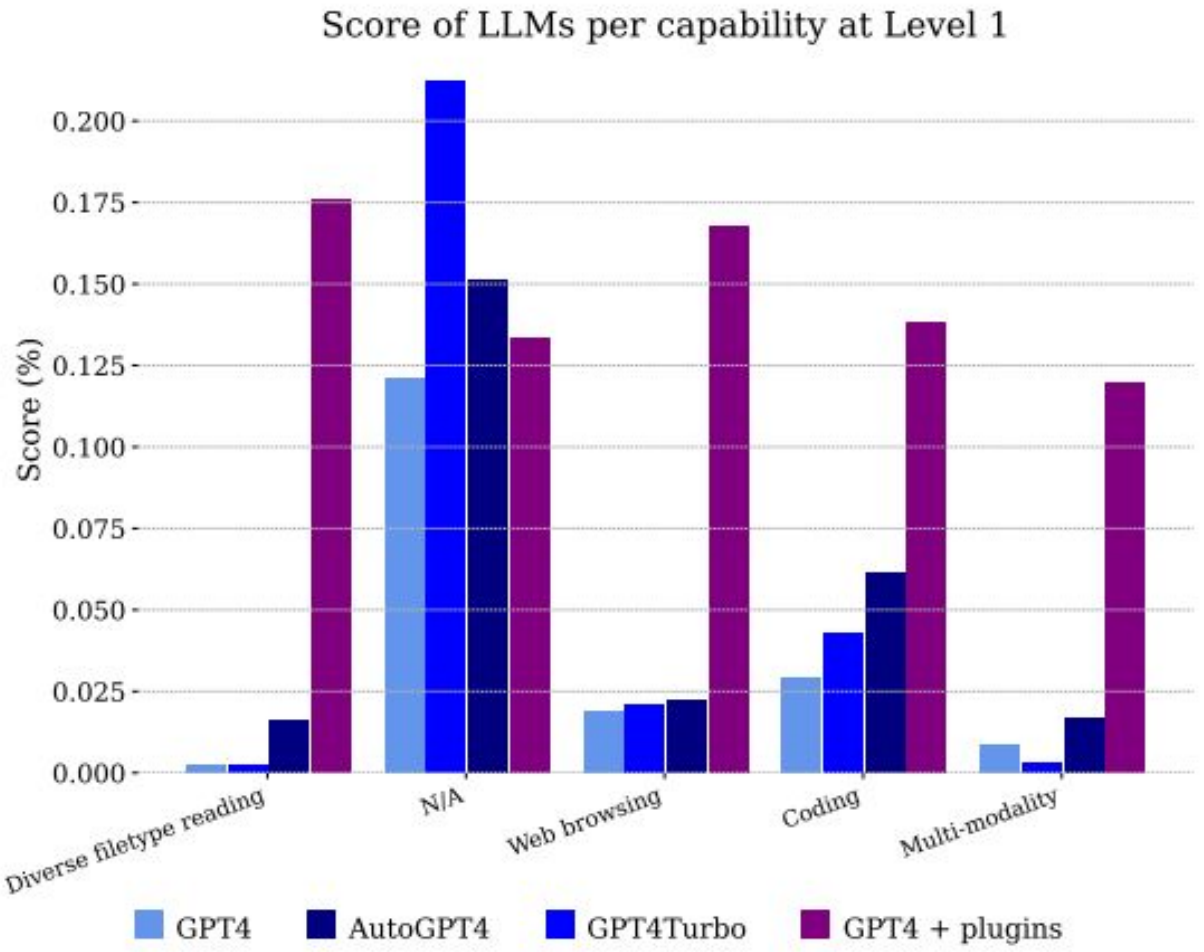
Overall accuracy on the original GAIA benchmark



Performance vs. Time to Answer



Which Capabilities Benefit from Tools?



GAIA · OFFICIAL TEST LEADERBOARD · NOV 2026

Have Modern Agents Caught Up?

Agent system	Model families used	Score
Keen AgenticOS V1.1	DeepSeek · Kimi · Claude	**94.35%**
AuraAi	Claude	**92.03%**
Human baseline (original)	Human annotators	**92%**
Harness V6	Grok	**90.37%**
GPT-4 + plugins (original)	GPT-4 · manually selected plugins	**~15%**

Modern agent systems are self-submitted; original baselines used different setups.

Sources: GAIA original paper and official GAIA leaderboard · Nov 7, 2026

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06 Discussion & Limitations

What Did GAIA Show?

Research question	Answer
How to evaluate assistants?	GAIA design
How capable are assistants?	Bad , worse than humans
Effect of difficulty?	Difficulty level affect
Effect of tools?	Tools sometimes help , depends on the type of the tasks

GAIA gives a shared test of system-level capability—not a diagnosis of which component failed.

Does Overall Accuracy Reflect Agent Capability?

WHAT THE METRIC MISSES

Equal weight, unequal difficulty	A Level 1 answer counts as much as Level 3; Level 3 is only 75/466 (16.1%) of the score.
Repeated-run reliability	An average score may hide which tasks the agent solves consistently across multiple runs.
Only the final answer counts	Correct output does not establish reliable reasoning; tool use and efficiency remain unscored.

Proposed metrics

1. Difficulty-balanced accuracy

$$\text{BalancedAcc} = \frac{\text{Acc}_1 + \text{Acc}_2 + \text{Acc}_3}{3}$$

Each GAIA difficulty level contributes equally, regardless of its number of questions.

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2. Consistent success across repeated runs

$$\text{ReliableSuccess}_k = \frac{1}{n} \sum_{i=1}^{n} \prod_{r=1}^{k} \mathbb{1}[\text{correct}(i, r)]$$

Fraction of tasks solved in every one of k runs (n tasks total).

What Does GAIA Actually Evaluate?

An agent may combine several components to answer a question.

Vision → **Browsing** → **Reasoning** → **Answer**

Identify objects | Find evidence | Combine facts | Final response

Limitation	A wrong answer does not reveal which component failed.
Authors' rationale	Evaluate the whole system, independent of its architecture.

Dataset Limitations

01 **Costly to construct**

Human question-writing, solution traces and independent validation require substantial effort.

02 **Residual ambiguity**

Even after independent checks, some questions or accepted answers may remain ambiguous.

03 **Limited diversity**

Linguistic and cultural coverage is narrow; results may not generalize across languages.

04 **Source dependence**

External websites and files can change, disappear or become inaccessible over time.

From GAIA to Gaia2

A shift from finding information to acting in changing environments

GAIA (2023)		Gaia2 (2025–2026)
Read-only questions	→	**Read + write actions**
Websites, documents, files	→	**Simulated apps: chats, email, calendar**
Static question + answer	→	**Events unfold while the agent works**
Final-answer accuracy	→	**Task / state-change verification**

800 scenarios · dynamic events · temporal reasoning · ambiguity · adaptation

Example: Who Should Order the Cab?

Message colleagues I’m meeting today to ask who should order the cab. If nobody responds within 3 minutes, order a default cab.

01	**Find today’s colleagues**	Read calendar and identify contacts
02	**Ask via ChatsApp**	Send messages and monitor replies
03	**After 3 minutes**	If there is no response, order the cab

Success depends on actions, timing, and evolving messages—not a final text answer.

Gaia2 Is Far From Solved

Gaia2-CLI · OpenClaw 2026.4.1 · Same agent harness

Model	pass@1	Time based task pass@1
Claude Opus 4.6	**57.0%**	**3.8%**
GPT-5.5	**56.4%**	**4.4%**
Gemini 3.1 Pro	**52.0%**	**2.1%**

Even leading agents solve only about half the tasks.

DISCUSSION · RESEARCH PROPOSAL

Proposed Future Research
- In-context learning (ICL) from human solution traces

Can few-shot human solution traces improve accuracy and efficiency on unseen GAIA tasks?

BASELINE		HUMAN-TRACE ICL
Same agent + tools Zero-shot instructions	VS.	Same agent + tools Few human solution traces ICL

Experiment	Use held-out GAIA questions; compare random vs. similar demonstrations.
Metrics	Answer accuracy · Tool calls · Runtime · Token cost

Conclusion

01 A benchmark for complete assistants

GAIA tests real-world questions with verifiable answers, independent of an agent's internal architecture.

02 A large original capability gap

The original study found ~92% human accuracy versus ~15% for GPT-4 with manually selected plugins.

03 Progress reveals new challenges

Modern agents have closed much of the GAIA gap; Gaia2 evaluates dynamic, time-sensitive actions that remain difficult.

From finding the right answer to taking the right action.